

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0511

Wixom 138 kV
July 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0511. The Interconnection Customer proposes a 240.2 MW wind facility interconnecting at the Wixom 138 kV substation in the Kentucky area. The request was received on August 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 202 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,874,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SKEENA to PIN HOOK SUBSTATION 138.0 kV L	ALBION RD to WIXOM 138.0 kV line	147.1	129	42.6
WIXOM to ALBON ROAD 138.0 kV	LEE COUNTY SOLID WASTE ENERGY to LIVERMORE 345.0 kV L	126.2	255	64.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SKEENA to PIN HOOK SUBSTATION 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,391,000	58% (\$3,111,000)
WIXOM to ALBON ROAD 138.0 kV	Rebuild line with higher-capacity conductor	\$2,593,000	68% (\$1,763,000)
	Total allocated to project		\$4,874,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.